# ActInf Livestream #029.1 ~ "Active Inferants: An Active Inference Framework for Ant Colony Behavior"

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looks good go for it hello everyone welcome to the active inference lab live stream number 29.1 today is september 21st 2021 and we are going to be discussing this paper active inference and active inference framework for ant colony behavior this is a recorded and an archived live stream so please provide us with feedback so we can improve our work we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at all these social media links and all backgrounds and perspectives are welcome here we will be following good video etiquette or livestream here at the short link you can find all of our upcoming live streams that are planned um and if you want to get in touch or if you want to participate please reach out to us and we will invite you to participate in a panel today our goal is to learn and discuss this paper active inference an active inference framework for ant colony behavior uh we will be maybe going through some of the aims and claims keywords figures uh and having a discussion around this paper so just to start off we will go around and introduce ourselves and then start off with some warm-up questions so my name is blue knight i am an independent research consultant from new mexico and i will pass it to marco so hi my name is marco i'm located in vienna austria i'm a postdoctoral fellow at the conrad lawrence institute for commission research and my interest is very much from the clinical perspective i have to do with dbs patients and i have come across active inference actually rather recently cool daniel thanks um i'm daniel i'm in california and as first author i'm happy to maybe just introduce the big picture um you wanna just briefly say hello and then i'll return stephen are you here steven yeah hi there so did you just introduce me hi stephen here um nice to be on and uh i am based in toronto and um i work with communities and work with spatial sense making around different issues and um how active inference can now help to understand those processes of sense making dynamic inference in in the modern world awesome so daniel do you want to maybe introduce the paper or give us a summary of what was your big takeaway message uh from this paper or what was your motivation behind the work well just hearing some of the backgrounds of those here ranging from human medical and clinical to sort of social and collective processes playing out in humans it just shows that we each approach active inference from our own experience so personally my background was in genetics and insect behavioral ecology and so it was conversations and reading the papers of especially axel and of course carl fristen and we were having conversations about where does active inference fit into discussions about collective intelligence where does active inference fit into bigger discussions about evolutionary theory like if active inference can be an integrative framework for different social systems for different human systems why not other animals too and uh so there we were at the intersection of curious about big questions in evolutionary theory which i know we'll return to today and specific questions about the areas that i was researching at the time which was the regulation of foraging and decision making in ants and so it was conversations with with axel and maxwell to a large extent where we started thinking how could we modify the types of models that we currently have in active inference and bring that into something that adequately at least starts to work towards capturing how the ants do it and alec had worked on the code of many active inference projects and so that was quite a fun process maybe we can also talk about like interfacing between ants and computer science we mainly wanted to navigate a little bit towards the bigger questions in biology but do so by really tackling a specific scenario that would then lend itself to unique explanations and predictions and models and the specific contribution that differentiates the active inference model from other active inference models is first that it's in insects and ants but also importantly that it's stigmatic so it's modeling explicitly the feedback between the environments and the agents whereas there had been sort of multi-agent active inference simulations where the agents communicated with each other or they had a shared theory of mind those types of models existed but this was the first time that one of the action affordances of the agents included the capacity to modify the environment and the status of the environment modified the likelihood of the agent taking different kinds of actions cool so i just want to reach out to um marco and steven and just ask like what's something that you were excited to talk about today or something that you liked or remembered about the paper either you have thoughts should i start so one thing as i already mentioned before is the role of dopamine which i find super interesting and you mentioned the paper that the foragers have higher levels of dopamine so i was wondering about the role of that and how that influences affordances and what the individual and also the level above like the the colony does with that so is there a specific role for dopamine for the for the colony as a whole great question and we wrote it down on slide 14 so we'll definitely return to talking about some of the molecular roles of dopamine cool and stephen do you have any thoughts yeah i suppose mine's a bit more of a broad question but i'll be really curious about that balance between sort of specific um things relating to ants and how that relates to things like dopamine and maybe the broader process of like the design thinking and working through this kind of active inference framework just how that was useful to think about how things may relate to their environment so i suppose the process of creating the model as well as what the model was and what it found out about things in the world thanks awesome daniel it's a great point um almost asking about how do we study the biomimicry not of just materials like light materials or strong materials but of collective algorithms and we can't really study the success or the attributes of a collective behavioral algorithm outside of its ecological niche and so a very long and important branch of research by my phd advisor professor deborah gordon which is where i was when we were doing a lot of this initial work she studied for 30 plus years understanding how the statistical regularities of the niche were then associated with which collective behavioral algorithms of ants we saw happening and so there's like 15 000 ant species there on every piece of land except for basically antarctica which is kind of where they should be if you think about it but if you want to have a framework for ant behavior that includes the desert and the jungle and all these types of different environments it sort of baked in this idea of appropriate generalization because like we definitely can't have a theory for each ant species so we need to look a little bit bigger than that yet we don't want to have just an aunt specific theory only and let's you know think about termites and bees and non-insects so what is the appropriate level of generalization and how are we going to generalize the study of collective behavioral algorithms and how they interface with different kinds of environments while also remembering that you know we got to have our eyes on the ground we need to be explaining those ants we can't just simply spiral off into the abstractions about collective behavior so they answer a cool system because they remind us about ecological and behavioral diversity and the need to generalize but also there is a real system that we can actually see awesome thanks steven yeah thanks daniel i was wondering how much um that generalizability that is in the field that you're in with your supervisor you know that might be something that really brings something to the table for active inference and i'll be curious as well you know how much those algorithms that have been worked on for a number of years either got modified or ported or had to be kind of reconceptualized from the ground up you know that'd be interesting cool daniel you have a response yep so one reason why active inference is able to swap in to where other theories or frameworks have been used before is because at the core it's an input output structure just like a lot of other behavioral studies like sensory information comes in something happens and then action comes out so in that way it's able to map onto a lot of other ways that people have thought about animal behavior or ethology one of the differences in active inference is that we're not committed to any specific cognitive model so like in this paper we just used pretty much the most simple possible cognitive model if you could even call it that it was one step decision making there was no memory there was no anticipation there was no capacity to use some of the sensory affordances that we know that ants actually have and use in nature like multiple sense interactions with nest mates interactions with other physical objects so we mapped onto this input output structure but we know that we can generalize it later but so that's one reason why active inference will um hopefully play a nice generalizing role there um i i hope yeah go ahead stephen yeah that's really cool and i suppose one thing that when you mention that i suppose this even ties in with that dopamine question but i'm not sure because i'm not as familiar with that but is by not having it all based on the cognitive it's showing what the niche can do for you in a way it's like how much you know we all assume it's all going on in the head which is the old way of thinking but with the active influence it's like well maybe there's other ways that we can get where we need to get so to speak awesome and with the ants it was almost never a question of it happening inside the head only because it's it's not to demean their cognition it's just very clear that no single forager would know how many seeds does the colony have how many does it need through time what's the weather and the trade-offs of foraging where are the seeds i mean those are questions that at a first pass we know are just unknowable and so it's pretty clear that the way that individual nest mates are making decisions has to have a radically different mode than for example let's wait till we have all the information and then make the well-ration decision based on our simulation it's just not happening and then once you see that lens on ants you go huh well maybe there's some other forming settings where the agents don't have total information and even if they did it wouldn't be obvious what to do like we can all imagine a chess board we have total information about where the pieces are but even experts in chess don't necessarily know the best move and there might not even be necessarily one perfect best move so it really moves us into a setting of thinking about how action is undertaken including action that can be epistemic like gaining knowledge amidst uncertainty rather than waiting until we've resolved our uncertainty because then what we still wouldn't know what to do cool that leads in pretty nicely to one of the questions that i had brought up last week so there are these colony level metrics right like how many seeds are in the nest or how many you know uh nurse nurse maids are there relative to infants or foragers relative to you know a baby yeah i guess so how are these colony level metrics retained in a colony good question well you asked a why question in a sense so it makes me think about the tin bergens four wise which we've spoken about um on other live streams but the evolutionary um why is just colonies that have certain colony level attributes are more likely to succeed and so we end up seeing the emergence of certain ratios of male to female nest mates or certain ratios of forage or to nurse performance just by virtue of their survival so that's sort of an evolutionary circularity just like in active inference we talk about how the systems are resisting dissipation so systems that are doing it it sort of explains itself they're here and we observe them because they're resisting dissipation it's really this question about colony-level phenotypes that fascinates people who study collective behavior because there's the traits that a nest mate has which are not so different than the traits that any other insect has like six legs what's the width of the head what's the gene expression of the gland or the mid gut like what digestive enzymes are they making those are things you can measure about one nest mate and so the body plan in ants is really quite similar to all insects but then there's this really qualitatively different level of organization with a colony and how do those phenotypes arise how are those colony level phenotypes in feedback with nest mate how does selection shape those phenotypes nice thanks for your answer um and there's a couple so maybe let's go to the um this slide that shows like where you're measuring the colony level metrics or or the the coherence right like the colony coherence or i forgot the term that was used um but you have the model or maybe maybe we should start with the model really i guess so so maybe do you want to just give us like a quick overview of like the model and the point of it and um yep let's go to slide 15 and so just to bridge from the colony phenotypes back to the specifics of the model it's sort of like if there was a music festival there's all kinds of phenotypes of the individual participants like their self-report where did they move how is their decision-making being influenced by their sensory stimuli but then there would be certain descriptions that you'd only have one of for the whole music festival like what is the average distance of people so that's what we were studying with the swarm coherence metric but it just points that there's going to be two totally different types of phenotypes some can be measured multiple times within a colony per nest mate and other times there's going to be phenotypes that are only arising at the level of the colony like the sex ratio or the average distance and what our model did was we specified the process by which the nest mates basically perceived and acted so it was an action and perception model at the nest mate level we didn't explicitly have a input or an output for the quality not that it couldn't be done you could also think about where that might come into play maybe in dot 2. but we looked at a setting that has been used to study colony decision making and then we specified a nest mate level model and put it in this context where people had already been using it to study the dynamics of how colonies make decisions so here it's called the t maze test and there's two arms to the tmas the two um red arrows on the left side here of figure 1a and food is on one side of the tea maze and it can either be static or it can be switching and this is something that they use in lab rat experiments as well so the colonies task is to basically get food what we did is we didn't actually encode that as like a colony level goal we really went down to the nest mate level and we specified just like any other active inference model what are the sensory observations made by nest mates which in this case was just the pheromone density of the squares around them what are the action affordances of each nest mate which is sort of it can move in any of the adjacent squares what's the generative model which entails a preferences and a free energy minimization happening inside the nest mate at each time point so that given the observations at a certain time the actions are selected and in this case the preference was for more of the pheromone so like the pheromone can be interpreted as a attractant something that the aunt pursues or prefers to observe and um the only other tweak which is what introduced the pheromone in the first place and the whole concept of stigma g was that on the way in after they've touched the seed but not before they also deposit pheromone so that was an additional action affordance to deploy pheromone and there wasn't any decision making about that it was just if you've already touched the seed then you lay down the pheromone and if not you don't and so it's sort of like those simple rules pursue regions of higher pheromone density lay down pheromone if you've touched a seed but not if you haven't those two rules at the nest mate level recapitulate some of the phenotypic dynamics of colony decision making so that's why it's fun to think about like maybe we can look at complex collective phenomena and then understand how they emerge from simple interactions among agents and agents in their environment nice stephen just a quick question is i'd be interested because you've got say one ant that goes off foraging type of thing and then maybe you've got the second one that follows and then i suppose there's what goes back to the nest if i'm right um yeah i was just wondering like how much more complex or does it help to have more in the modeling or does it make it more challenging to start using too many ants so we templated the natural history if you will of this off of a formica ant species um like a wood ant so it's referenced in the paper this dynamic of like go out without leaving pheromone come back with pheromone so given that natural history to change the number of ants from two to two thousand to two million it's just a number in the config file when you run the code so it actually is not challenging to scale the colony's size it might burden your computer or cause some other limitations and it certainly will change the collective outcomes maybe in a linear or maybe in a non-linear way like we explored in the figure three there are so many natural histories and ways that ants forage like there's tandem running or like an ant remembers where the food is and then it does like a leap frog and it takes an uninitiated ants nest mate and leaves it and then when those two return they bring back four and then there's eight so there's all kinds of recruitment and food localization strategies in ants which is why it's important that there's not just like one single way that ants forage because they're looking for things that have different regularities in the within a natural history the script is written in a way where it's pretty easy to change the map shape or to change the number of ants or their appearance dynamics and then a little bit of the challenge would be like to adapt it or transpose it to a different natural history or to add multiple interacting pheromones or other kinds of things like interactions amongst the ants marco yeah sorry i just wanted to ask about the the the model as it is described in the paper because um at the beginning the ants have basically random movements right because all the surrounding step movement opportunities are the same uh probability right so is it sort of like a browning motion until the first ant discovers the the food source and then starts altering the environment by the pheromone placement that's exactly right and there's seven supplemental videos gifts where it's shown and what it kind of looks like on this team ace it looks like you see these little particles that start swelling and diffusing up from the bottom so that's a total brownian walk because there's no preference for direction when there's no pheromone deposited you see one ant find the seed and then it starts laying down this pheromone but it's also diffusing randomly because at least initially there's also no preference for direction but then you can imagine that there starts being a higher concentration of pheromone on the side with food than not and eventually that gets reinforced because it starts to attract anyone who sort of diffuses into this regime of a locally increased pheromone density they start to get pulled to the places where there's increased density and then that ends up reinforcing and as soon as there's like a ridge of high pheromone connecting the food back to the nest entrance that's the formation of a pheromone trail so no ant needed a bird's eye view or a goal to even form a trail it's a purely agent-level process-based explanation by which you get the emergence of a phenotype that goes beyond but one nest mate knows or even prefers and that's the existence of a trail now you can imagine that colonies that have the appropriate parameters like the right amount of explore versus exploit how much to follow one pheromone how rapidly should the pheromone decay those parameters are actually shaped by the evolutionary process colonies with maladaptive decision-making of nest mates are not going to bring home the seeds they're not going to reproduce whereas for any given niche if the regularities can be captured by the decision-making algorithm that the ants use you'll see those parameters plus slight variations on them in the future awesome thank you and that kind of brings us to a figure two which i um threw up here and i i clicked this quote from the paper so it's talking about the rate of pheromone decay and it says informationally this decay rate can be thought thought of thoughts of an environmental perturbation or drift that counteracts the reinforcing effect of stigmatic convergence and so in this and here even in the figure legend it says like we chose the representative simulation did not turn the parameters to discover optimal decision making um and so i was just wondering if you played with the decay rate at all because i think about like if the pheromone didn't decay there would be a really difficult time switching like from one arm to the maze to the other when the food source switches so you have to forget at some point where the food was then this kind of brings me to one of the big questions is like how important is this decay rate in like the memory of anthony like so this is clearly like an example of niche modification where they lay down the pheromone and it externalizes their memory of where the food is for other ants in the colony but there's this um you know it has to decay otherwise they're just confused like if we were flooded all of the time with all of our memories like unable to distinguish our our memory from the present time which happens in some forms of you know uh neuro neuro like diseases narrow atypical people have this like you know inability to distinguish the present from the past from the future so just is that like a critical thing or do you have any comments on that maybe or maybe stephen wants to comment on what i said and then uh you comment or stephen do you want to go first or what do you think okay yeah i'll just add one thing to that um yeah i think that's a a good point is i suppose how much does the ability to allow the environment to re i suppose re-remember i'm not really remember but reformulate so you may not need to forget as much as just change how much you're weighting what you you um give priority to so in a way it's like by knowing something that you're more certain about it might get something similar so i'll be interested about that idea of you know offloading onto the environment and how that maybe negate some of that anyway yes so it is um akin to forgetting but what's really interesting is it's the colony for getting because the ant doesn't have a direct memory so as we encoded it so it's a lot like cumulative culture there's a totally qualitative difference between what an individual remembers on board and then what the culture remembers and so we do modification on our cultural niche breeding scott and so that enables new dynamics of learning and forgetting which are distinct have different mechanisms than the ones that happen at the nest slash individual person level and so it is um a forgetting that occurs and otherwise you would get stigmatic reinforcement stigmatic tightening and it would end up just being over fit to the way things were and then you wouldn't be able to access new resources or have the flexibility to find the seed after it moves to the other branch sorry did you have something else to add how goes it scott hi scott um so here we'll switch on to oh steven go ahead soon yeah just one thing daniel i was wondering how much you think i mean i know it's an ant model but you know you're talking about extrapolating to beyond how like you just mentioned that with culture so that um i wonder if there's some principles that starting to emerge that you think you're kind of curious about exploring further because of that ability to think differently about how we know the precursor model to active inference was actually on slide 53 the acquisition of culturally patterned attention styles under active inference so this was a visual scanning visual foraging model where the eyes scanned over pottery designs and there were cultural preferences for certain local motifs it's kind of interesting that we almost stripped the cultural interpretation away to apply it to ants and i do think that it could be re-integrated in it just helps us clarify that we can know what the individual level mechanisms are like of personal preference what is being sensed by the person and what are their action affordances kind of like all the variables in the active inference model perception cognition and action and then the external states can be other agents and we've talked about communicating agents before but the external states can also be something that's cumulatively modified and so when your sensory capacities include the ability to read websites and then your action affordances include the ability to modify websites it would take transposing the model into those different inputs generative models and outputs but it does suggest that we don't have to make everything only existing as a function of interactions amongst active inference agents in fact cumulative culture helps us distinguish pieces that are clearly cognitive clearly on board from outcomes that are clearly cumulative and have memory that isn't just what the agent remembers almost gives a a rationale for fashion in a way could be interested in as well when you think about design and of pottery there's a fashion that there's there's a fashion of other things as well that that makes me think about like who controls how the seed switches arms so here it's the spatial movement of the seed from one side to the other and the switching dynamics but fashion i mean i guess what people wear or just like oh it's a fashionable book it's sort of a way in which people's niche modifications include locating and moving or hiding seeds and it's like oh you didn't know about this band like i foraged better because i know this band that you don't know that's sort of like hipster informational foraging and so how do people signal through fashions you know keep up with the time cubism is the art that we all look at this is what we think is cool or this seed location is no longer in vogue so maybe some types of foraging metaphors could help us understand cultural phenomena that also feels like hope that also feels like irving goffman kind of reference of projecting out as you want to be perceived that those recursivity so that you you know if you want to be perceived as a mountain climber you wear a shirt from a mountain climbing store um and so you're you're expressing a thing and you want to be perceived as that so there's that kind of that mirror effect that goes on um uh to define ourselves by how others see us it also reminds me the earlier notion of the cumulative embodiment reminds me of the david bone work that all reality is first a thought and so the realities that are accumulated are just accumulated that then become artifacts of that interaction and then you know their embodiment is left in that intergenerational communication through language and through material culture is where the mind resides really in the interaction with those things so we're able to communicate in deep time that way they it reminds me that quote that bach and mozart didn't die they turned into their music right so we still communicate with them we still hear them at who they what they were communicating out the realities that they were trying to create became embodied and they still available to us through deep time and over space as well and so that maybe part of the cultural affordances are the leveraging of the communication the leveraging of self and identity and you start to have people able to pick up on those and and leverage themselves and then they take it and take it in a slightly different direction then you have that innovation that happens and one of the things i find fascinating is in fashion and in food are two places you do not have intellectual property protection of copyright but they're the most innovative or if you're from england in innovative areas which is fascinating right because they don't need the excuse for intellectual property laws the rationale is that it helps to promote innovation but in fact the most innovative things don't aren't afforded that protection they're more in a foraging kind of um enthusiasms that make those things you end up with the intergenerational maintenance not just construction and enshrinement of a pyramid but the actual ongoing maintenance of whereas it's not enough to just do it once and then leave it behind and so it is interesting about how process and carrying the process forward gives you the static objects of cumulative culture but the cumulative culture and the static objects themselves they don't map one to one onto a process let alone adaptive one or a successful one and so how do we and that relates to um the way that mod that reward was modeled in this um simulation and also to dopamine so i know we'll get there awesome uh stephen and then scott this reminds me as well i was that there was a marketing manager that i once spoke to and he was telling me what they they pay people to go into bars and to drink certain drinks in trendy clubs in like the big cities say in london or whatever so that they would be seen to be drinking that drink and then it starts to spark off of course it's completely covert but this sort of thing you can see how that distorts people's um stigmatic trail because they see someone who's presumably very good looking or fashionable or very high status or whatever at the bar sitting on the stool drinking the drink in a conspicuous way and they they take the cues at some level so anyway i thought that's interesting for sure okay so i think we can maybe move into oh sorry oh sorry scott i was looking i was looking for a shoot can you hear me okay sorry i was looking for the controls i totally lost my screen to sort of screwed up um the end in the u.s stephen there's a thing i was looking for the word for it but there's actually when you go into a bar and say i'd like an absolute vodka please if you've been hired by absolute you're required to say i've been hired by absolute vodka i'd like an absolute vodka please they actually have a disclosure requirement for that exact behavior where people go in bars and if you're being paid you need to make a disclosure as a technical matter under under federal law so i thought that was pretty interesting um there was another point i was going to make but i spaced on it because your point i'd like your point so much sorry it's like an active um disclosure here's my preferences here's my expectations here's some components of my generative model right yeah i was just wondering since uh since you were talking so much about the distributed cognition now and making sort of extrapolations there there clearly must be some limitations there right because i i assume or guess that like human culture societies have a larger capacity to change phenotype compared to let's say the limited variables that an ant colony might have and that might be subject to uh evolutionary pressures and and then is there let's say is there is there a larger um flexibility and phenotype for for human culture would you say almost have affordances that include changing the hyper parameters the higher order parameters on our niche and clearly the scope of our niche modification has expanded as well so it'd be like if this ant colony you know we added another level it can decide where to move the seed to or it can decide to modify the structure of the maze i mean we know that ants build they build their nest architecture in many cases so what happens when you enable a new kind of performance it reveals like a whole new game theoretic or signaling theoretic dynamic that can flip how it would have been otherwise like in a fixed maze maybe there's a strategy that works but maybe if you add this ability to move blocks maybe the strategy is actually to um you know make a hypotenuse just going straight to the food so that totally changes which parameters of learning and forgetting you might want to have and so it's a very interesting question about how affordances come into play how do they interact what is happening with humans with our new affordances our ability to have long-range synchronous conversations and to make irreversible ecological decisions as well those affordances for niche modification and communication are changing what it means to be human definitely agree with that um as we kind of mentioned in the dot zero contextualization video any more thoughts on that or we can maybe move on to these colony level metrics in figure three figure three just to describe the two on 23. um on the left is just a tally of basically the seeds returned home just the count of the round trips that is being shown through time steps on the x-axis for four different numbers of foragers like 10 30 50 and 70. again we weren't like tuning parameters or making statistical claims but it's interesting to see that the 30 50 and 70 the ones that kind of look like step funk same accumulation dynamics as the others do and then on the right side is that swarm coherence the um the distance metric amongst the nest mates and so again maybe this could be run out or it would be different on the map shape and all these other things but just to see that the average distance sort of increases and then stabilizes at a somewhat characteristic rate so these are phenotypes that nest mates don't have but they're composed of the phenotype the position the behavior etc of the nest mates and so again it just points to the need to be clear like when are we talking about individual level phenotypes that might be in feedback with environments or with collective outcomes and when are we actually talking about specifically collective outcomes nice so i have a question that um you know relates to this directly so in this kind of colony cohesion or i think you call it form coherence um there's the distance metric that is measured but but in real ant colonies like are there other metrics that kind of are used to judge this kind of cohesion because i wonder about like the time that's spent communicating like ant communication time like does this also like enhance colony performance or what other kinds of metrics can you like use to determine this kind of cohesion we talked about this a lot with the authors and we so we thought okay well there's like the euclidean distance the average distance or the distribution of distances amongst estimates one could also look at the correlation of their trajectories so a perfect correlation of the trajectories would be like every ant is taking the same trip whereas no correlation of trajectory would mean like every ant was going on a different direction or having different movement dynamics and so that could be computed over the total trajectory of each ant or even at each time step like on the next time step zero correlation means like that there's 50 going up 50 going left 50 going down 50 to the right or maybe there's like this sort of shift a coordinated shift within a time step as well as among time steps so looking at the correlation in trajectories the mutual information um these kinds of things could be calculated for the colony and then we thought you know it might not even be easy to rank these because what if the mutual information is low because they're doing different types of foraging and that is constituting a more coherent swarm so just because their correlation is high that's sort of like first pass oh well it must be a coherent swarm because everyone's taking the same trajectory but then there's this other type of coherence that might even be more emergence when the correlation or the mutual information of trajectories is low and it's actually by virtue of that that the colonies able to achieve things that the nest mates don't awesome thank you stephen yeah can i just ask when you mention that could that um about the the different types of movements and correlation um do you think that could point towards um how there could be different types of counter factuals brought to the table so as maybe the information coming back as well as being a seed it could be a counter factual um i i suppose the war it's not a seed but it's a wall or um that i don't know maybe that doesn't mean i haven't thought it through but i'm just wondering how counterfactuals might play out founder of factuals is it's an interesting question because it's a it's a mental ascription which we think we can do but then do ants do counterfactuals or how do they at least act as if it's a great question just one thought would be the resilience of their algorithm is such that they act as if they're taking counterfactuals into account so for example their trail is functional when it's functional but then it's always like this adjacent possible that the trail could be disrupted and when the trail is disrupted they find a way to go around or to repair the trail and so it's like by having this counter factual like the adjacent possible that they might not have a functional pheromone density there's this self-repairing and regenerating aspect to their behavior in nature that makes it look as if they are accommodating these counterfactuals and it reminds me of one of the greatest disinfo scenes of all time in a bug's life i believe or maybe ants like where leaf drifts down and then all the ants are carrying the seed of course in their hands not in their mandibles which not how ants carry seeds and then like the overseer comes over and blows the whistle everybody stop you know the first ant that was returning home gets anxious it's like you couldn't stack i know it's just a movie so i know chill out etc but you couldn't stack more misconceptions into that representation of how resilience actually arises in foraging in overseer doesn't come and the first ant engages with the barrier and then goes around it or over it and that is what ends up allowing the rest of the ants who don't have communication with any overseer or even that first ant to participate in that re-functionalization of that part of the trail so it would be cool to see representations that don't disinform as to how ants actually work that's really cool that's a really cool response actually i also suppose you could flip that as well and say well maybe the adjacent possibles of these paths that the ants are taking in the environment is maybe that's how we get our counterfactuals in the brain it could be like a foraging path you know um i'm not saying that's how it is but maybe at times there's something that mirrors that but in in our own cognition scott you know that daniel that reminds me of our misunderstanding of how ants work it reminds me of the assertion i read that dogs in the wild don't have the alpha dog kind of characteristic and the impact mentality i don't know i don't know the research on this but that they learn that from humans that and so it's interesting we kind of project onto animals human ideas about stacking or whatever social status and it's fascinating to think whether in art because we were talking about fashion before might the arts be a vehicle for inviting humans to start to understand things that ants know right you know what what can we find from the art art world a movie or something like that which did a different treatment of it and started children embracing that right because these are children are learning from these movies it's a fascinating notion to think about how a more i use the word accurate portrayal but a more maybe sensitive portrayal to the dynamics of other biological beings might help us as biological beings to find our way because certainly the abstractions that we've we're teaching children now are not fully functional this so there may there's some additional learnings there the conversation i was on just before this we were talking about um capacity building in precarious populations and i was pointing out a similar kind of thing we were saying well we might have things to learn from precarious populations it might not just be parachuting in colonial type solutions but rather what can we learn from new discoveries and this was in the context of um the idea of bringing technology to for information to precarious populations but in the reverse of it bringing community knowledge to technologies the reverse flow and again similarly here maybe the community knowledge of needs to be brought to the technology architectures that we're talking about rather than us constantly being blinded to the architectural elements that we might have because we have these centralized elements these alpha dog notions these things that we project out so we're unable to learn because we're just imposing a structure on something that may not be there so we can't learn nice daniel the history of psychological projections onto the ants and the bees is tremendous it relates to sex and gender and politics i mean she's called the queen and she's called a she but she's just a diploid insect so how do we go from the words that we use and the kinds of models that we use to describe how the colony functions is it an economic maximizer is that the goal is gdp maximization or is getting seeds the goal and that was one of professor gordon's insights as well is like it's not just about maximizing the per capita return on foraging in terms of seeds which optimal foraging theory often suggested because in the desert it's costly to go out foraging you are spending water to get seeds which give you energy and water but it's kind of like there's a time where you don't actually need to get more seeds so the way that the projections occur and that they are mapped sometimes implicitly or explicitly to our human systems and and ideologies is very interesting and i think that relates more to this general point in active inference which is actually that sensory observations are ambiguous that's the insight of helmholtz and unconscious perception which is like the perception of ants even if the edge detection is relatively unambiguous what it is that the ants are doing whether they're taking care of every nest me and it's like utopia or they're sloughing off the useless ones and it's like hellish that's our projection and so that's why it's so interesting to have a model where we can be clear about what the sensory observations are of ants of of ourselves and then how our generative model is actually what is generating perception you know we see with our mind not with our eyes so the ant is also perceiving with its mind not just its antenna awesome stephen that's a really very very good point and also i think with helmholtz bringing that back in is really useful in terms of how we project because he was a way for us to get back to what we already knew it's like with all our intelligence we as psychology came in and this knowing and this perspective that knows and that becoming this almost unspoken axiom that we didn't even know was unspoken everything from there on in was about what do you think about this and what do you think about that and what do you think about the other and 100 years later you know we're in all sorts of problems right um well were very quickly in all sorts of problems actually at the beginning so suddenly yet you know going back before then i'd be interesting to know about 300 years ago they probably look at us with a wry smile and they'd be like yeah right because they hadn't had a whole load of other stuff involves this knowledge of knowing as being about thinking and us and it was much more in the kind of you know unknown states of the world i mean their dictionaries weren't even there right so they didn't have that defined language in the same way um so yeah i think that's that's a nice point well lang language becomes oh sorry scott and then daniel language becomes that artifact again that's why i keep uh yeah again it's not it's not accurate what i say when i say the mind resides in language and material culture but it's i just keep saying it to get people off thinking that it resides in the brain and so it's it's just those traces we leave of prior interactions that seem functional become so ascendant you're born feral and then you kind of look and observe what works and it's so localized knowledge is so localized and now it's become the access to broad knowledge is is so great that there's nothing that suggests that the human lifespan was designed to be able to absorb all the knowledge that's now available right there's no we're not necessarily able to do it but we try i certainly try i got i'm surrounded look at all the all of us that have books around us you know we're like we're surrounded by so we're trying to figure all this stuff out but there's nothing that says we should be able to and certainly not as individuals right so then you then i start thinking well okay i'll take solace in the fact that we this is the synthetic intelligence thing i keep you happy about instead of artificial intelligence we know so that's good so that's pretty nice so that means that what i got to do is just cultivate the we not the i so that jump is what it feels like i'm wondering it just it feels like humans could be better served by at least being sensitive this we're not going to become ants right but we would be better served by a sensitivity of this and it feels like they there's a vast amount of human de-suffering that could be made available if if this could be made available and i'm not sure if it's through the arts or through economics or what the mechanisms of uh influence will be to allow enjoyment of the additional solution space or additional existential space that the these observations offer you know it doesn't necessarily change the way we act every day but it definitely could adjust the existential aspects of the human condition and that seems like they would indirectly serve to reduce anxiety and maybe hatred as a result anyway sorry for the ramble daniel stephen you brought up um 300 years ago and so it wasn't exactly 300 years ago but it's within about five to ten years of 300 years ago was a book called the fable of the bees private vices public benefits i think it's it's worth reading one of the summaries of what this poem and book is about um the thesis is that contemporary society is an aggregation of self-interested individuals necessarily bound to one another neither by their shared civic commitments nor their moral rectitude but paradoxically by the tenuous bonds of envy competition and exploitation so it's a bit of a parable private vice and public benefits in a bee colony and about how and why that succeeds or it doesn't let's work with the preferences and sensory capacities and affordances that we have and then think about how to engineer that with stigma g and cumulative culture to the kind of collective outcomes that we want to see rather than try to optimize it at potentially a lower level or without the whole picture in mind which could lead to not just um private vice but public vice as well yeah yeah thanks um so in a way what you're saying when he's when they're talking about envy and such like it's more like um i suppose phenotypical or social property rather than this individual oh you've got it well done so right but now we we've tend to now focus as an behavioral abnormality i suppose or and you lose that capacity or we've lost that capacity or and hopefully now we're gaining it to speak at that bigger level and not just the bigger level in terms of a statistical social science averaging but one where people are actually still purposefully engaged in so that mid scale type of thing so daniel and then i'll give my thought oh yeah blue oh so my um that kind of ties back to what scott was saying about um you know like all the books that you have around us and also into this like concept of synergy and it's something that i was thinking about like just yesterday we had this literacy report come out in new mexico that was just horrible right like the kids can't read i mean not for a long long long time but maybe you know it made me think like we are doing this method of communication here like via youtube and connecting like over support we're into this like oratory or even like i don't know audio visual like is there a is there a big bigger word for that like a video transduction of information and it's like are we you know maybe the need to read and write is becoming obsolete i mean as we're moving in and as we're you know externalizing all of this into our niche environment i mean in the history of early humans history was an oral tradition and then it became a written tradition and now maybe the need for reading and writing is is not there because we're back into this ability to directly communicate across vast distances and to leave recorded traces of this audio visual in time i mean yes in a media that's less permanent i mean if the internet survives or whatever but i mean we're all under the assumption that you know in 50 years if humans are still alive this video will be archived in the wayback machine or somewhere right and so maybe are we are we losing the need for that i don't know just a thought ah damn good so i'll give a comment on that related to ant brains and then we can go to the dopamine discussion so there's um a scaling relationship that people have observed with social mammals where the larger the group size of the mammal the larger the relative brain is so that suggests that sociality in mammals includes like memory the ability to simulate counterfactuals potentially theory of mind or thinking through other minds like we talk about in active inference but being social for mammals is clearly like a cognitive load and in the ants a trend in the other direction is observed where larger colony sizes have often smaller relative brain size of the workers and that's one of the key differences between social and youth social so a lot of times people will talk about the bees and the ants as being social as if they're hanging out or doing it because they like each other but the reality is that ants have been colony living for 120 million years of evolution slash one creation however you want to look at it and they're not doing it just to hang out so it shows that once you make this phase transition from social which might have increased cognitive burden to you social it might be associated with decreased cognitive burden and decreased cognitive necessity of the individuals because it's like it's good to follow the trail and go off on your own and just wander around and leave the trail when you find new food etc those are simple rules that aren't necessarily requiring tremendous onboard cognition yeah scott maybe that's why um populations of humans don't have to be educated maybe we're going back to that and say maybe we don't need high levels of teaching for everybody in society there will always be there's always going to be a distribution of education and many dimensions of that education all we can hope is that the cumulative context supports everyone i mean the reality is even even if we tried not everyone's going to be a brain surgeon and a rocket scientist and an ant scientist and this and a lawyer and a doctor and a grocer right so we have to specialize and parse and so it's kind of interesting because the aspiration of self cultivation suggests that you want to have a you know strong body strong mind and every individual needs to be the representation of everything but it's not maybe that aspiration of power being residing in an individual through individual capacities is not scalable you know even though it's observed in mammals at a certain scale maybe at a certain scale you have to be used social because the intrinsic complexity of the society and the information available is such that it's impossible to reside in a single brain but the again one of the things i say is that i tell people the brain is infinitely capacious because it's just an antenna and it's just tuning into the mind which itself has whatever capacity every capacity so again maybe we need to train more not in substance but in process of learning or in knowledge discovery rather than in facts so that people can navigate the mind that's collective with their brain that's an antenna i have a similar thought actually scott like maybe i mean we're we're becoming connected like you know in even just 100 years ago we were very dispersed and so maybe as we're becoming more connected we're becoming more like a high having developing more of like a high mind where like the reliance on survival or cognition is is less so stephen and then we'll go to daniel and ant brains i'm just going to say just in instead of just in time manufacturing is just in time thinking and also tying into that process piece um you know how much it is about getting the knowledge i either book off the shelf and how much is it knowing the policies for how to engage that book can be the thing and that's where it can be valuable to see an old video ought to have met you know someone who actually met einstein would have known that other side to how he wrote those books right we just have this encoded form and there's other ways things can be encoded right um and that may be what you experience when you have a chat with someone maybe a professor or someone who wrote the book which is very there's there's there's that quality to it and that that might be um something where thinking about because in a way these ants really they've got to have trust in their action policies scaffolded by some sort of pheromone trail and we in if you actually look at most psychiatry is basically taking the individual and trying to maybe reduce their anxiety or the distortions changing serotonin or dopamine levels and i think what this is showing is well you know that might make someone less anxious and less you know less challenged in their integration into society but there's another there might be other ways and that speaks to eco psychology and other things which maybe seemed a little bit hippie before but they can be scientifically rigorous from what we're talking about and that's great so just a quick little interject question because steven brought it up and this is something that i've been like thinking about and i was gonna bring it up with with the counter factual point but like is there a role for trust in the social insect synergy like like can you maybe daniel has examples of ants that are liars or ants that can't be trusted or like is there some kind of like deviant form of ant thought or behavior and then what it brings i'll give one interesting example of that which is um under the clean division of labor model one would expect that queens should just be like super fertile and workers should never have any capacity for fertility and so when it was found empirically in honeybees and in some ants that some workers had partially activated ovaries like they retained the capacity to reproduce which is also facilitated in those insects because unfertilized females can make male eggs just by making a haploid egg initially that was framed as cheating and it was in the um the game theory of cheating and all of these ways that you could think about economic exploitation free rider problem but at the same time it's the case that queens do die and when that happens the call in me is almost hedging its bets against one failure mode by having this just-in-time capacity to salvage some reproductive fitness with pre-activated workers so was it deceptive at one level was it adaptive at another level there's many examples like that and i would say it all falls under that ambiguous stimulus heading i'll ask a question from the chat and then we can because it will take us towards the brain um sergio asks how can the autonomy of an agent for example an immune b cell living within another autonomous meta agent like a mammal be accounted by colony behavior and stigma g it's an awesome question it speaks to the nested formalisms so ants also have mobile cells and we're abstracting around that when we just have a nest mate level model but potentially nested modeling could speak to that because t cells um are perceiving their local niche and then they're they're acting in their niche they're releasing cytokines and they're um changing the immune context which could bring other t cells or b cells down and then i posted a paper distributed adaptive search in t cells lessons from ants with professor gordon and others that was explicitly about that it was like a triple play collaboration with ants immune cells and computational search algorithms and the way in which computational search algorithms and computational frameworks like active inference although not in that paper give us a common descriptive ability across the ants and the t cells but then we once we frame it we learn from the ants and the t cells and the richness of nature and that expands our computational models so it's like a feedback in the informational niche between the kinds of cognitive and computational models we propose that helps us generalize and then we get surprised by ways that nature actually works out that feeds back into how we modify our models so maybe anyone else can give a comment on the the b cell and the ants and then otherwise we can go to the dopamine you think we're good yeah to go to the ants in the dopamine actually marco i was curious how i know it's a big question but how has dopamine been seen in the human neuro community and clinical side and where does act inference come into play or expand on that because i suspect that we might see some parallels with what it's doing in ants as well but what is the situation been and where is it going with human dopamine well i think that depends on who you're asking there is for sure one big uh focus on uh the rewarding effects or let's say the role of dopamine in addiction um and you know just as as a positive prediction error signal um but then of course you have so and and that you might put at the um you know at the dopamine uh tract that comes from the vta and goes basically throughout the the cortex and also to the nucleus accumbens but then you have also the the movement facilitatory role of dopamine which stems from the substantia and innervates the striatum and basically serves a action policy selection role via selecting competing affordances i guess you could say and and by that uh you select one um movement uh strategy and if uh you lack dopamine then uh basically you um you're freezing in your movements and that's one of the cardinal symptoms that you get in parkinson's disease and as an overarching principle i guess at least that's the intuition i have right now is that dopamine somehow modulates uh how some one like an agent interacts with the environments it's it's more this um this interface how how do you weight uh signals from the environment that you can react to and and that's i think the aspect that combines the the rewarding function of dopamine with the motion and action um permitting function cool i see a lot of interesting parallels with the ant world so first for those who might not be familiar ants and invertebrates have a lot of the same molecules in their brain as mammals do so like some of the same exact neurotransmitters like serotonin and dopamine which is why some of the same drugs that mimic the structure of those neurotransmitters have effect in invertebrates as well as in vertebrates but they don't have the same anatomy of the brain so like you mentioned the vta and the substantia the ants and the invertebrates don't have those exact structures however they do have some structures that play similar computational or functional roles so point one is that's an argument for why we need to look beyond just the brain layout of the because not all cognitive systems inside or even beyond animals have that exact layout so it motivates just like we've been talking about like looking at the specifics really understanding how does these different dopaminergic signaling pathways work in the mammals but also what are they doing functionally and computationally so that we can understand and design other systems and then um dopamine has been studied for playing a role in the regulation of individual foraging in invertebrates as well as in vertebrates so people have done experiments with giving dopaminergic drugs to insects and seeing how it increases their foraging activity and even in the ants that are um shown on like slide 14 those ants the ant going into the nest entrance there is painted because it's been treated with dopamine so it's possible to collect ants and then either treat them with dopamine and measure how much it influences their dopamine levels in the brain we found that increases in brain dopamine through pharmacology increased foraging activity and we found that providing an inhibitor of dopamine synthesis decreased ants foraging activity relative to like a control group on the same day so on first pass that is consistent with this reward role for dopamine and that's the kind of empirical observations that people have found confirms their belief that dopamine is playing a role in reward but then what you said at the end there that reward and action come together with sensitivity and so we can actually add in something that we know about these ants and about other ants to think about how does dopamine play that role well in this ant species we know that individual nest mates make the decision to go foraging or not probabilistically depending on the rate of interactions that they have with incoming successful forager nest mates so if there's a high rate of incoming successful foragers it stimulates more foraging and if there's a low rate it sees this foraging activity it's like if you block incoming foragers from coming in there's a lag followed by a drop off in foragers leaving and if you have an excess of foragers coming in it leads to an excess going out which eventually does deplete the workforce so it's not like there's infinite ants they can just keep on sending but that's a self-regulating process so that when things are good and food is coming in send more it's a good day to forage but if there's only a trickle or there's like a predator blocking one of the paths then no more ants go out and so one of our hypotheses although it would take a few other kinds of experiments to really go into this in detail would be that dopamine plays a role in modulating the sensitivity to interactions of nest mates with each other and their sensitivity to changes in environmental conditions and so by framing what dopamine does as modulating precision or modulating sensitivity to interactions we integrate the mechanisms that we know actually underlie the regulation of foraging in ants with the empirical outcomes that increases of dopamine signaling increased foraging activity and decreases decreased foraging activity uh so stephen and then marvin i'd be interested where your thoughts are in terms of the type of foraging whether it's kind of an anxious like foraging if that makes sense because that could be because often um i think that some of the serotonin inhibiting uh medications they're helping to reduce anxiety and there could be a difference between the way people are reading those sensory inputs so i suppose could be could the ants be also foraging more if they're anxiously foraging if that makes sense maybe a sudden peak and then at some point they they there's that that doesn't become such a dominant dynamic for addressing reward do you want to respond daniel and then we'll go to marco yes there hasn't been a huge amount of work on serotonin forging but there is one paper i'll post in chat which is serotonin modulates worker responsiveness to trail pheromone in the ant phytoledentata so it is known that different neurotransmitter systems play a role in modulating the sensitivity to interactions and whether that gets psychologized or projected again i don't think that's always the right move to make but empirically serotonin does play a role there so that makes me wonder is um so the the neural modulators that you usually find in in mammal brains do they all exist in uh in ants as well like uh including um norepinephrine acetylcholine etc so it's a very interesting question there's some that are kind of like a level celebrities in vertebrates and invertebrates like dopamine and serotonin then epinephrine and norepinephrine which are very important for the regulation of arousal in vertebrates their functional roles are to a large extent played by actually the neurotransmitter pair of tyramine and octopamine in invertebrates which are trace amines received at the trace amine receptor in vertebrates and so the exact proportions of these different neurotransmitters differ in vertebrates and invertebrates but the metabolic enzymes in the same gene families and that was some of the most interesting analyses was actually to look at the sequence evolution and the duplication and deletion events of the metabolic enzymes that transform neurotransmitters into each other from common amino acids into neurotransmitters synthesizing and degrading them and then the protein receptors that are used on the cell membrane like the dopamine receptor so the gene families are homologous the chemicals are similar and um neuropeptides there are some that are ancient gene families that are shared across animals and there's other vertebrate specific and invertebrate specific neuropeptide gene families so they're more similar than dissimilar and there's a massive amount of overlap and that's why again like drugs that are designed for working in one species often have a similar effect like neonicotinoid pesticides they affect the acetylcholine receptors of invertebrates broadly just like nicotine affects the acetylcholine receptors of humans right um yeah i think the the really cool um promise of active imprints is that you can understand different cognitive systems in terms of the messages that are being passed and there are lots of comparisons you can make and i guess that's also a good um general test of the of the substance behind the whole framework to um to try and make those mappings where you say you have signals that get passed from bottom up like typically by a glutamatergic um trains of spikes let's say they might be matched by expectations that come from top down which might be inhibitory and then you have all sorts of uh modulatory uh signaling molecules that's the precision of certain messages there so and and once you have that framework established you should be able to find it in all kinds of different cognitive systems whether they are structured like the brain where you have physical pathways that link one region and another or whether they are let's say [Music] more based on externalized elements like uh pheromone trails or i mean very much at the beginning but i think there's lots to discover yeah marco i just want to respond to that really quickly you know we were i don't know if you caught our last live stream i think it was 28.2 just the one that was last week but we were talking about um you know it's interesting we're talking about mental wandering like what does mental wandering have to do with like foraging or ants but there's a lot of a lot of connections to be drawn and really like we're thinking about um you know tracing a thought through the brain like like which is i mean my background is also neuroscience which is like passing a message from one neuron to the next neuron and really the basis of the paper was mental wandering and meditation and like the you know not suppression of thought but like when you consciously inhibit you know the thoughts from wandering around so it says mindfulness practice it's really interesting because in the ant colony there's so many like relationships that i can see like i mean even between like foraging like when you're wandering and you're mental wandering is that like thought foraging or but i mean also more than that because there is that like synaptic release and so is that i wonder if that like synaptic release or synaptic transmission that happens in neurons is similar to this like pheromone deposition right so like you have synaptic release and that encourages more neurons to speak signaling so you have this pheromone deposition and that encourages more ants to keep foraging on those trails and so i really do wonder how much of the what our role is in neurotransmitters is externalized in this kind of niche modification and external cognition which would happen for us internally like but i wonder if the answer are doing that as part of this like high mind mentality um sorry steven i i was rambling yeah i know it's very very interesting points one thing i really picked up as well and i um with what you're saying in marco is this message passing that you have in active inference it is very liberating because you traditionally talked about in terms of signals and signals ends up maybe being electrical signals and it ends up being like a wire that we think of in the you know other like electronic world and and it's a very big trap you know so then there's somehow a signal going through and then next thing you know you you know everything implicitly or explicitly becomes a computer but this this just this general idea of message passing somehow so signals still it's hard not to make signals about you know a transmission system more like in a in a wire but message passing it at first i used to wonder why they're using that term because it seems so ambiguous and like what's going on with that but in some ways i can see is so much benefit from that from that that formulism yes the signal and signal transduction signal processing which is frameworks that are used in electronics but also we talk about like signal transduction in the cell so it shows how wide the usage of that signal term is and it almost implies an essentialism to the meaning of the signal like it actually carries a meaning but when we think about message passing we can be clear about what that message is look at how the generative model interprets what is the incoming stimuli or observation instead of treating the signal as what is bearing meaning we can think about the stimuli as just being the empirical observation which as we've been discussing from the sensory level on up through the narrative observations are ambiguous and they require a generative model in order for sense making to occur and so agreed that not only is message passing away that bayesian models can be updated like we learned about from bert devries and others but it's also a way that we can step away from a sort of signal versus noise partitioning paradigm and move into an empirical observation meeting generative model meaning maker marco yeah uh so what i also like about this but i'm i might um have a completely different perspective on this than than uh you know you have but um the the the nice thing about messages is that uh it takes you away from this object orientation where you identify things and you can say they either are okay or they're malfunctioning but you rather look at the interaction between let's say vertices or objects and and you find maybe more illuminating answers there in if you think about psychopathology for example so that you can point to one brain structure and you say ah that brain structure is broken and that's why i have symptoms it's rather uh to think about it in terms of well maybe this message cannot be transmitted and uh and that's uh maybe the reason or or that's where you have to look for answers uh so one other thing about message passing that's i think a great way to think about active inferences it really grounds it in information theory because that's like shannon information is is like you know spawned from that idea of message passing and coherence and a message but also in the active inference framework it's always subject to interpretation uh stephen yes i um i was as i started to think as you were saying that marco about someone shaking hands with someone else okay so you transmit a handshake you pass a handshake you know i mean because both people have to reach out whereas when you transmit a signal it can be a one way a more than one way thing i suppose in some ways i think i i sort of agree what you're saying blue i also would say in some ways that shannon entropy is probably a bit stronger on the signal side so in some ways it it gives that effective message passing but active inference goes further down well it seems to offer further down that route it gives this um other way to extract so it's not like the noise is diminishing the signal um although it's it's there's other ways of getting that through action um what's in the signal um so i suppose in a way it's a bit like the handshake it's like what's that hand saying to me the funny handshake from the mason well i reach out and i hold it you know and i get a bit of a better feel and i see what i want to pass up my uh predictive processing chain it just makes me think about how when we think about signal occurring amidst noise it's almost like we're getting one unit of stimuli whether it's rgb channels on the tv or whether it's a frequency range for audio or whatever it happens to be if the task is to find the signal in the noise we're always going to get less than one could be 99 signal to noise it could be 5 so we're kind of always going less when we think about the stimuli as being a message that's passed to us and something that we enrich we can make it more than one so that secret handshake or that shape can be unpacked and understood in a broader context so for example how many you know bits of information does it take to communicate how to make lasagna or something it's like you don't need to describe every single sequence somebody who knows it only takes one bit of information maybe just the letter l or something like that signal processing again makes us think about diminishing and filtering out what we're getting in whereas maybe active inference helps moves us towards understandings where there's an enrichment of the information coming in which is always required but especially when there's sparse or ambiguous sensory stimuli nice steven that's a good point daniel i mean that sort of takes you to the value of varied stat fairy tales and yeah the idea i hadn't really thought about that before but the idea that the generative model is free to make more of something um and embellish it i suppose and of course that that may also really come into its own when you're bringing in multi-sensory information so you know there's there's this uncertainty but there may have to be something which embellishes um early on maybe that explains why children early on get very into dreams and uh very wacky sort of modeling of the world um they maybe want to extrapolate beyond you know what that sound and that light source and that feeling and that smell of food is you know the porridge for the three bears that goldilocks is eating is is actually going to signify for them so yeah that's interesting did we already have this like conversation about do ants dream of undead crumbs did we have that conversation daniel is there a book or something right i think you dreamed it okay okay so that brings me to my next question was it you know i wonder like i i just was um listening to josh steinbaum's lecture and i read his dream code dreamcoater paper in which like the computer explores like potential recombinations of symbols and stuff in like a it's like an off phase like a helmholtz machine there's like a wake sleep cycle in this algorithm that um he was using and so you know i mean i wonder i guess ants sleep i mean they like rest do they rest they have to like assimilate and update their generative model do they believe people have studied circadian rhythm and the foragers have a strong circadian rhythm again probably not every single species but we did some work with gene expression and found gene expression cycling of circadian rhythm genes in the brains of foragers but not nurses in this species so nurses always indoors low-key working all the time they take a lot of micro naps and foragers have more like phasic circadian rhythms especially for insects that use the sun to navigate like honeybees well so what i really wonder is what about the collective um like ant generative model right like you think about the the this is like a collective behavioral or collective consciousness like is there you know dreaming or potential like recombination like exploration does that happen in like an ant colony it's an awesome question and another example of um true colony phenotypes and people have studied the activity waves in ant colonies over multiple time scales like a minute time scales as well as the circadian rhythms and that is actually not dissimilar from the multiple bands in the eeg so just like the electrical dynamics of the brain which present as one sort of just time series you can do a decomposition to extract out like the alpha and the beta and the theta bandwidths so just like there's only you know one 2.4 gigahertz wi-fi wavelength yet there's multiple wi-fi networks that can co-exist so how does that multiple coding happen for the signals that we make like wi-fi and radio for the brain which is uh using different frequency bands simultaneously and there's also these oscillation dynamics in ant colonies so if the whole colony and this a wave like what i mean what kind of signal technology are you using to detect these waves i have to i'm not similar to the way that the time series of the eeg you would do a sort of uh four-year decomposition to look at the different frequency modes similarly people look at the spatial position of ants and then can ask what are the activity dynamics through time and one paper i'll post in the chat is from 2016 by gel all emergent oscillations assist obstacle negotiation during ant cooperative transport so actually having this sort of like oscillating group activity helps them not lock on too specifically to an object's direction that it's being pulled like if there's some object and then there's the nest entrance yeah okay we'll pull in a straight line isn't that the shortest distance between two points yes but first off no one knows exactly where the nest entrance is and second off even if it were the most direct it might not be the most resilient or tractable but if you just say okay you know we're gonna pull a little bit this way and zigzag it this way and zigzag it that way you get that wisdom of the nest mates because they each have a different estimate on exactly which way to go they get averaged out and if there's a little block somewhere built into the movement that this object is being brought home with is the ability to go around it and so that's another example about how processes that are dynamic at the very very micro level with the agents often constitute more adaptive emergent processes and more tractable ones it's like we either get the resilient tractable agent level rules or we get the fantasy well we should just move it in a straight line everybody should know exactly where to go and if they don't we're going to punish them it's not how it works awesome um and so that kind of leads nicely into i think we were talking already actually about like the wisdom of the crowd um and the example with the ox maybe that was in the dot zero or maybe last week but um it really leads me to thinking about um collective behavior and collective intelligence and just um that might be like a nice question for people to give thoughts on because i i just it's in my own interest to find out what other people think about the difference between collective behavior and collective intelligence and where is um you know any thoughts there on that stephen sort of linking that and what you're just saying now with what daniel was just saying in some ways the brain so if you've got going more chaotic which i suppose in some ways you can do through your environment or just and the environment will do it to you and then you've got this synchronization of waves as was mentioned so you could go from that kind of synchronization of waves and i suppose it could become more coordinated or it becomes more chaotic so you've got this kind of balance in terms of sense making between more entrainment between more coordination of action either between agents or between the sensory motor properties of the organism or which would in some way be more focused you would imagine and has that risk of rigidity but has the advantage of efficiency um so brainwaves seems to have a bit of a bit of a mid you know this this synchronization of brainwaves gives you a sort of a mid-scale potential so it's not completely chaotic but it's not completely you know synchronized yeah i don't know i'm not sure what the difference is really between collective intelligence and collective i um there's this cool diagram collective behavior and collective intelligence and it shows cooperation and coordination as essential to collective intelligence but but it also then like that those are both behavior cooperation and coordination and then what's left of it's cognition intelligence or like what is cognition anybody have any like thoughts on i'm not marco i know your uh background is neuroscience so let's let daniel think and then i'm going to put you on the spot studying collective systems it always struck me as interesting because the organism is a collective of cells and so it's more like an approach that we take to any system to see it as arising from interactions at an even lower level and so when i look at these very human centric thought maps informal thought maps it includes things that you know open source software p2p business those are human-centric and so it ends up muddying the waters and introducing a lot of psychology and game theory and terms that aren't like distinct like is cooperation distinct from a coordination and that's one reason why having a perception cognition action model even though it comes across as very austere and distant from the phenomena that we want to explain like coordination cooperation and cognition having such an austere but scalable and flexible model helps us get at these terms without presupposing them as natural things that exist in the world so we get to just model what is on the ground happening like with the ants or on the ground in another context and then ask how our definition or parameterization or formalization of coordination and cooperation are coexisting as descriptions of the process that we're specifying at the actual proximate mechanism level nice stephen and then marco yeah i think that that challenge of these things having a psychological or folk psychological or cultural labeling is is limiting and by stepping back from that might which in some ways makes creates more uncertainty i could say that it seems more austere but it's more austere because it is less clear initially but the benefit of being less clear and this actually ties in a bit with some of the infusion space work that i've been working on is it makes it more nascent it um whilst with active inference it gives that holding container so by being more nascent you've got more ability to then proceed again with another type of naturalistic potentially more naturalistic framework and not being tied to a naturalistic based on language and all the distortions that can come from that so i think that that's a really big part of what makes it useful oh sorry no thanks um these are all super good points um and i agree uh completely and i think that this is also recognized in um in psychology that there is a tendency to attribute all kinds of uh terms to uh different brain areas for example um like the prefrontal cortex there are more functions assigned to that it could possibly hold for example you could find papers that identify the location of um stock trading behavior and these are clearly human invented terms and there is no reason why they should have um an expression in the in the brain um and so um i think uh a nicer way of thinking about this and um i think you mentioned it before is uh this dynamic between segregation and integration so sometimes it makes sense to be focused on a particular task everyone let's say uh without in lots of distractions you you are very efficient at performing one particular task um but that might not be the most creative mode of operation um where you might rather want to have lots of different brain regions simultaneously interacting with each other and exchanging lots of information that's not efficient but it's very integrative so um neither of those extremes is probably viable by themselves but if you have a good flexible way of moving between segregation and integration you can be very much task oriented specific but you can also be creative and incorporate lots of different information it's a good point marco and also that meta task of appropriately choosing how to shift tax tasks is framed as a type of mental action in the nested active inference the deep active inference formalism so we can actually think about that shifting between different affordances at the attentional level as another kind of action selection task that uses this same perception cognition and action framework so and i agree as well stephen it does start with a more atomic germ it's more nascent and that's what allows us to to map and bridge and compose if we try to put too much meaning into the lowest level we actually constrain how tall the buildings that we can build are and douglas hofstetter talks about this in the context of reflexive computer brains and ant colonies just today as it was in 1979 three of the most important systems to understand and gets at in all of those systems how the lowest level like the interactions among nest mates themselves or the letters they cannot have meaning in the way that for example words do because if you load all the meaning of words onto phonemes you can't compose sentences and it's just it's very interesting how even though of course we want that meaning we have to actually take a dip into using pieces that are sub meaningful in order to have flexible meaning because if you just try to run straight for the meaning you're going to end up with a very constrained system that might have the one function or meaning that you want to encode but it won't have the ability to reflect or to generate novel meaning so just in the last few minutes um i just want to thank daniel for taking the time out to discuss his awesome paper with us which i've been really wanting to discuss for several months um and hopefully he'll be back next week with more conversations about active infrared um and marco welcome it was really nice to have you here uh and so just if anybody wants to give a final thought on this paper or some stuff that they're excited to talk about next week well thanks a lot this was really and i learned a lot there are also lots of references i think that i need to check out now that you mentioned um and yeah really thanks a lot for for the paper and for presenting today yeah just to mirror what marco says and also welcome marco thanks for joining us really really enjoyed the the different avenues and actually the ability for it to keep coming back to something which could hold those different branches which is quite nice so we saw an ability to come back to and reintegrate a lot of different ideas so that was really nice so let's look forward to keeping that going well thanks everyone for joining and blue for excellent facilitator and broadcaster role i think for 29.2 what might be awesome other than having some novel nest mates on the stream and in the live chat what might be cool would be like to lay out some of these analogies that we've been exploring like okay agent and environment all right that's nest mate and pheromone density there's like the environment that you can control and then there's sort of the abiotic environment or there's the parts of the environment that are outside of the affordance of niche modification like the thunderstorm okay so maybe having that kind of a table would help us align a lot of the systems that we've been talking about so ants computers brains societies other systems that people care about that's kind of like us preemptively foraging maybe laying down the inkling of some trails because just like a few years ago when i was deep in the ant game seeing that there was a way to bridge active inference and ant research in potentially a new way maybe someone's going to go oh i mean architecture i do that every day and i've been looking for a way to inroad active inference or another another way that maybe we could help scaffold somebody else's research because i think the stigmatic modeling and a lot of these discussions go to a lot of systems and things that people are already working on so that would be awesome to talk about in 29.2 and to build momentum going forward awesome well thank you again everyone and we were are looking forward to our discussion next week so get in touch if you are watching on the live stream and want to participate bye